



Barker College

**2007  
TRIAL  
HIGHER SCHOOL  
CERTIFICATE**

# Mathematics Extension 2

Staff Involved:

- WMD\*
- JM
- VAB
- BHC

30 copies

**General Instructions**

- Reading time – 5 minutes
- Working time – 3 hours
- Write using blue or black pen
- Board-approved calculators may be used
- Write your Barker Student Number on ALL pages of your answer sheets
- A table of standard integrals is provided on page 12

Student Number .....

AM TUESDAY 31 JULY

**Total marks – 120**

- Attempt Questions 1 – 8
- All questions are of equal value
- ALL necessary working should be shown in every question
- Start each question on a NEW page
- Write on one side of each answer page
- Marks may be deducted for careless or badly arranged work

**Total marks – 120**

**Attempt Questions 1 – 8**

**ALL questions are of equal value**

Answer each question on a SEPARATE sheet of paper

**Marks**

**Question 1 (15 marks) [START A NEW PAGE]**

(a) Find  $\int \frac{(\ln x)^5}{x} dx$ . 1

(b) Use the table of standard integrals to help find  $\int \frac{dy}{\sqrt{4y^2 + 36}}$ . 1

(c) Let  $\frac{2x+2}{(x-1)(x^2+1)} = \frac{A}{x-1} + \frac{Bx+C}{x^2+1}$

(i) Find the value of  $A$ ,  $B$  and  $C$ . 2

(ii) Hence, find  $\int \frac{2x+2}{(x-1)(x^2+1)} dx$ . 2

(d) (i) Prove that  $\int_0^a f(x) dx = \int_0^a f(a-x) dx$ . 2

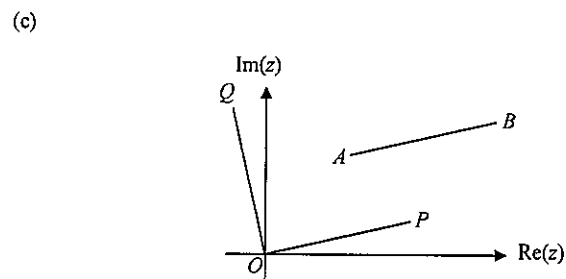
(ii) Hence or otherwise evaluate  $\int_0^1 x^2 \sqrt{1-x} dx$ . 3

(e) Evaluate  $\int_{\frac{\pi}{3}}^{\frac{\pi}{2}} \frac{dx}{1+\cos x}$ . 4

**Question 2** (15 marks) [START A NEW PAGE]

(a) Find  $\sqrt{21+20i}$  in the form  $x+iy$ . 3

(b) Sketch the locus of  $z$  described by the inequality  $|z-1+i|\leq 1$  and state the minimum value of  $\arg z$ . 2



In the Argand diagram above, intervals  $AB$ ,  $OP$  and  $OQ$  are equal in length,  $OP$  is parallel to  $AB$  and  $\angle POQ = \frac{\pi}{2}$ .

(i) If  $A$  and  $B$  represent the complex numbers  $3+5i$  and  $9+8i$  respectively, find the complex number which is represented by  $P$ . 1

(ii) Hence find the complex number which is represented by  $Q$ . 1

(d) If  $z = x+iy$  ( $x, y \in \mathbb{R}$ ), find and describe in words, the locus of the points  $P(x, y)$  such that  $\operatorname{Im}\left(z + \frac{1}{z}\right) = 0$ . 4

(e) Write  $\sqrt{3}+i$  and  $\sqrt{3}-i$  in modulus/argument form. Hence show that  $(\sqrt{3}+i)^{10} + (\sqrt{3}-i)^{10}$  is a rational number. 4

**Question 3** (15 marks) [START A NEW PAGE]

(a) The equation  $x^3 - 3x^2 + 6x - 4 = 0$  has roots  $\alpha, \beta$  and  $\gamma$ . Find  $\alpha, \beta$  and  $\gamma$ , given that one root is  $x = 1+i\sqrt{3}$ . 2

(b) (i) Prove that if polynomial  $P(x)$  has a root of multiplicity  $m$  at  $x = c$  then  $P'(x)$  has a root of multiplicity  $(m-1)$  at the same point. 2

(ii) The equation  $8x^3 + 4x^2 = 2x + 1$  has a double root and a single root. Find the double root. 3

(c) If  $\alpha, \beta, \gamma$  are the roots of  $x^3 - x - 1 = 0$ , find the equation with roots  $\frac{1}{1-\alpha}, \frac{1}{1-\beta}, \frac{1}{1-\gamma}$ . Hence state the value of  $\frac{1}{1-\alpha} + \frac{1}{1-\beta} + \frac{1}{1-\gamma}$ . 4

(d) If one of the roots of the equation  $x^3 + ax^2 + bx + c = 0$  is the sum of the other two roots show that  $a^3 - 4ab + 8c = 0$ . 4

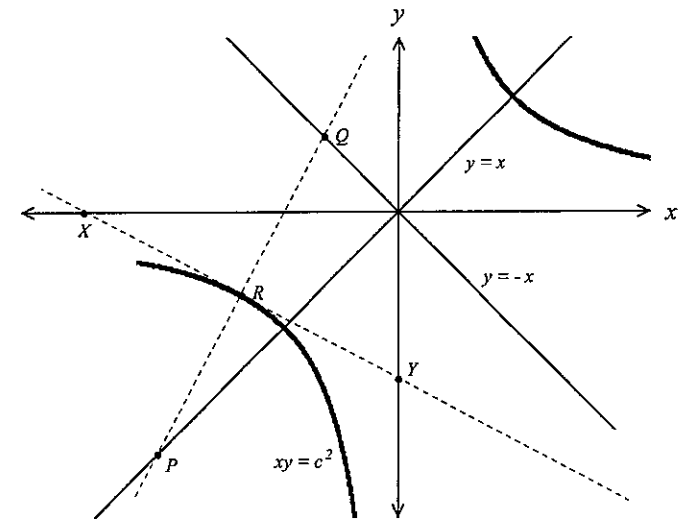
**Question 4** (15 marks) [START A NEW PAGE]

Marks

- (a)  $\phi$  is a complex cube root of unity, i.e.  $\phi^3 = 1$ ,  $\phi \neq 1$ .
- (i) Find the value of  $\phi + \phi^2$  1
- (ii) Prove that  $(a-b)(a-\phi b)(a-\phi^2 b) = a^3 - b^3$  2
- (b) (i) Sketch  $y = \frac{x-2}{x+1}$ . Show all points of intersection with the coordinate axes and any asymptotes. 2
- (ii) Use your graph in (i) to sketch:
- (a)  $y = \left(\frac{x-2}{x+1}\right)^2$  2
- (b)  $y^2 = \left(\frac{x-2}{x+1}\right)$  2
- (c) By considering the behaviour of  $y = \cos^{-1} x$  and  $u = e^x$ , or otherwise, draw the graph of  $y = \cos^{-1}(e^x)$ . Calculus need not be used. 3
- (d) Sketch, without using calculus,  $y = x|x-2|$ . Hence, or otherwise, state for what values of  $k$ ,  $x|x-2| = k$  has 3 solutions. 3

**Question 5** (15 marks) [START A NEW PAGE]

(a)



In the diagram above  $R(ct, \frac{c}{t})$  is a point on the rectangular hyperbola  $xy = c^2$ .

The tangent to the hyperbola at  $R$  meets the  $x$ -axis at  $X$  and the  $y$ -axis at  $Y$ .

The normal to the hyperbola at  $R$  meets the line  $y = x$  at  $P$  and the line  $y = -x$  at  $Q$ .

You are given that the equation of the tangent at  $R$  is  $x + t^2 y = 2ct$ .

- (i) Prove that the equation of the normal at  $R$  is  $ty + ct^4 = t^3 x + c$ . 2
- (ii) It can be shown that  $P$  is the point  $\left(\frac{c(t^2+1)}{t}, \frac{c(t^2+1)}{t}\right)$ . 3
- Find the coordinates of  $X$  and  $Y$ . Prove that  $Q$  is the point  $\left(\frac{c(t^2-1)}{t}, \frac{c(1-t^2)}{t}\right)$ .
- (iii) Show that  $PQ$  and  $XY$  bisect each other. 2
- (iv) Show that  $PQ = XY$ . 2
- (v) What type of quadrilateral is  $XQYP$ ? Justify your answer. 1

Question 5 continues on page 7

## Question 5 (continued)

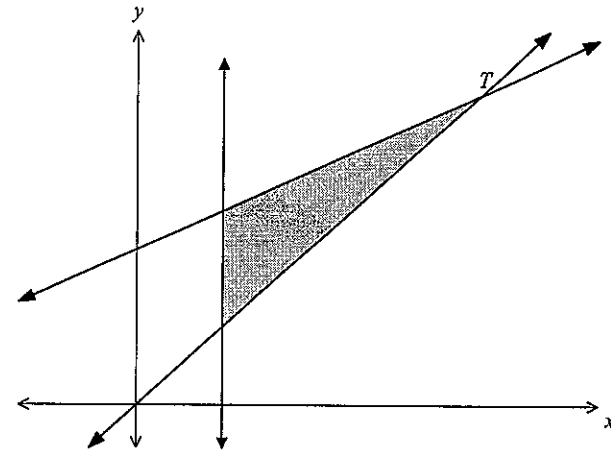
- (b) A tangent to the hyperbola  $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$  at  $P(a \sec \theta, b \tan \theta)$  meets the nearest directrix at  $Q$ . If  $S$  is the nearest focus, prove that  $PQ$  subtends a right angle at  $S$ , i.e.  $\angle PSQ = 90^\circ$ .

You may assume that the equation of the tangent to the hyperbola  $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$  at the point  $(a \sec \theta, b \tan \theta)$  is  $\frac{x \sec \theta}{a} - \frac{y \tan \theta}{b} = 1$ .

5

## Question 6 (15 marks) [START A NEW PAGE]

- (a) Find the Cartesian equation of the ellipse  $x = 1 + 3 \cos \theta$ ,  $y = 2 \sin \theta - 2$ . 2
- (b) The area bounded by the lines  $2x - y = 0$ ,  $x - y + 4 = 0$  and  $x = 1$ , shown below, is to be rotated about the  $y$ -axis. 4

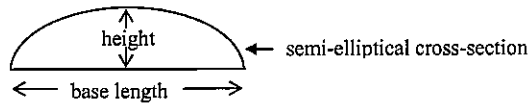


Show that the  $x$ -coordinate of  $T$  is 4. Then, using the method of cylindrical shells, show that the volume of the solid of revolution generated is  $18\pi$  cubic units.

Question 6 continues on page 9

Question 6 (continued)

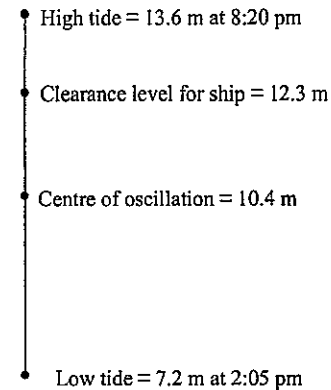
- (c) (i) Explain why  $\int_0^a \sqrt{a^2 - x^2} dx = \frac{\pi a^2}{4}$ . It is not necessary to use formal integration. 1
- (ii) Using (i) and symmetry, or otherwise, show that the area enclosed by the ellipse  $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$  is  $\pi ab$  units<sup>2</sup>. 3
- (iii) The base of a particular solid is the circle  $x^2 + y^2 = r^2$ . Every cross-section perpendicular to the  $x$ -axis is a semi-ellipse. The base length of each semi-elliptical cross-section is 4 times its height. 5



Use part (ii) to find an expression for the volume of a cross-sectional slice and hence find the volume of the solid.

Question 7 (15 marks) [START A NEW PAGE]

- (a) How many positive integers  $n$  are there such that  $n + 3$  divides  $n^2 + 7$  without a remainder? 3
- (b) Four families each have four children. What is the probability that exactly two of the families have two boys and two girls? 3
- (c) From the letters of the word RENEGADE, three are taken at random and placed in a line. 1
- (i) How many different 3 letter sequences are there with exactly one E in the sequence? 1
- (ii) How many different 3 letter sequences are there altogether? 3
- (d) The depth of the water in a harbour is 7.2 m at low water and 13.6 m at high water. On Monday, low water is at 2:05 pm and high water is at 8:20 pm. A particular ship requires water to a depth of at least 12.3 m to leave harbour. The ship's captain wants to leave harbour between noon and midnight. Find between what times the ship can leave, assuming that the motion of the tide is simple harmonic. 5



**Question 8** (15 marks) **[START A NEW PAGE]**

**Marks**

(a) (i) Use De Moivre's theorem to express  $\cos 3\theta$  in terms of  $\cos \theta$ . 2

(ii) Use the result to solve  $8x^3 - 6x + 1 = 0$ . 3

(iii) Hence deduce that  $\sec \frac{\pi}{9} \sec \frac{2\pi}{9} \sec \frac{4\pi}{9} = 8$ . 2

(b) (i) If  $I_n = \int_0^{\frac{\pi}{2}} \sin^n x \, dx$  then show that  $I_n = \frac{n-1}{n} I_{n-2}$ . 4

(ii) By first finding  $I_5$ ,  $I_7$  and  $I_9$  in unsimplified form, and noting a pattern in your answers, 4

show that  $I_n = \frac{\left\{ \left[ \frac{1}{2}(n-1) \right]! \right\}^2}{n!} 2^{n-1}$  if  $n$  is odd.

**End of Paper**

**STANDARD INTEGRALS**

$$\int x^n \, dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \quad \text{if } n < 0$$

$$\int \frac{1}{x} \, dx = \ln x, \quad x > 0$$

$$\int e^{ax} \, dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax \, dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax \, dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax \, dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax \, dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} \, dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} \, dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} \, dx = \ln \left( x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} \, dx = \ln \left( x + \sqrt{x^2 + a^2} \right)$$

NOTE:  $\ln x = \log_e x$ ,  $x > 0$

## Question 1

$$(a) \int \frac{(\ln x)^5}{x} = \int (\ln x)^5 d(\ln x)$$

$$= \frac{(\ln x)^6}{6} + C$$

$$(b) \int \frac{dy}{\sqrt{4y^2+36}} = \frac{1}{2} \int \frac{dy}{\sqrt{y^2+9}}$$

$$= \frac{1}{2} \ln \left\{ y + \sqrt{y^2+9} \right\} + C$$

$$(c) (i) 2x+2 \equiv A(x^2+1) + (Bx+C)(x-1)$$

$$\text{when } x=1, 4 = 2A \Rightarrow A=2$$

$$\text{comparing coefficients of } x^2: 0 = A+B$$

$$0 = 2+B$$

$$\Rightarrow B = -2$$

$$\text{when } x=0, 2 = 2 + (-1)C \Rightarrow C=0$$

$$\therefore \underline{A=2, B=-2, C=0}$$

$$(ii) \int \frac{2x+2}{(x-1)(x^2+1)} dx = \int \left\{ \frac{2}{x-1} - \frac{2x}{x^2+1} \right\} dx$$

$$= \underline{2 \ln|x-1| - \ln(x^2+1) + C}$$

$$(d) (i) \int_{x=a}^{x=b} f(a-x) dx = \int_{u=a}^{u=0} f(u) du$$

$$\text{let } u = a-x$$

$$\therefore du = -dx$$

$$= \int_a^0 f(u) du$$

$$= \int_0^a f(u) du$$

$$= \int_a^0 f(x) dx$$

-2-

$$(ii) \int_0^1 x^2(1-x)^{\frac{1}{2}} dx$$

$$= \int_0^1 (1-x)^2 x^{\frac{1}{2}} dx$$

$$= \int_0^1 x^{\frac{1}{2}} (1-2x+x^2) dx$$

$$= \int_0^1 \left( x^{\frac{1}{2}} - 2x^{\frac{3}{2}} + x^{\frac{5}{2}} \right) dx$$

$$= \left[ \frac{2}{3} x^{\frac{3}{2}} - 2 \times \frac{2}{5} x^{\frac{5}{2}} + \frac{2}{7} x^{\frac{7}{2}} \right]_0^1$$

$$= \frac{2}{3} - \frac{4}{5} + \frac{2}{7}$$

$$= \underline{\underline{\frac{16}{105}}}$$

$$(2) \cos x = \cos \left( \frac{x}{2} + \frac{x}{2} \right)$$

$$= 2 \cos^2 \frac{x}{2} - 1$$

$$\therefore \int_{\frac{\pi}{3}}^{\frac{\pi}{2}} \frac{dx}{2 \cos^2 \frac{x}{2}} = \frac{1}{2} \int_{\frac{\pi}{3}}^{\frac{\pi}{2}} \sec^2 \frac{x}{2} dx$$

$$= \frac{1}{2} \left[ \tan \frac{x}{2} \right]_{\frac{\pi}{3}}^{\frac{\pi}{2}}$$

$$= \tan \frac{\pi}{4} - \tan \frac{\pi}{6}$$

$$= 1 - \frac{1}{\sqrt{3}}$$

$$= \frac{1}{3} (3 - \sqrt{3})$$

$$\begin{aligned}
 & \int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \frac{dx}{1 + \cos x} \\
 &= \int_{\frac{1}{\sqrt{3}}}^1 \frac{\frac{2}{t^2+1}}{\frac{1+t^2}{1+t^2}} dt \\
 &= \int_{\frac{1}{\sqrt{3}}}^1 \frac{2}{1+t^2+t^2} dt \\
 &= \int_{\frac{1}{\sqrt{3}}}^1 dt \\
 &= \left[ t \right]_{\frac{1}{\sqrt{3}}}^1 \\
 &= 1 - \frac{1}{\sqrt{3}} \\
 &= \underline{\underline{\frac{1}{3}(3-\sqrt{3})}}
 \end{aligned}$$

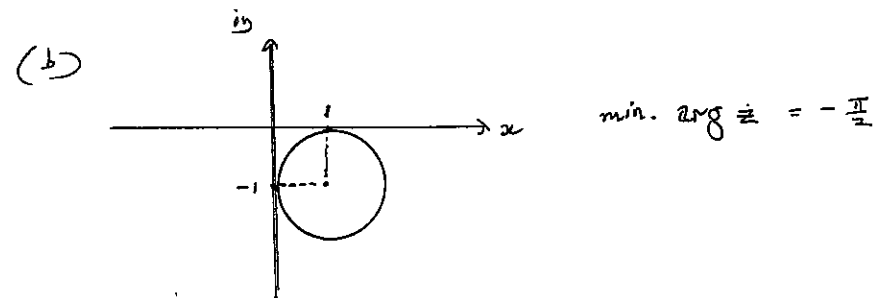
$$\begin{aligned}
 \text{Let } \cos x &= \frac{1-t^2}{1+t^2} \\
 t &= \tan \frac{x}{2} \\
 \frac{dt}{dx} &= \frac{1}{2} \sec^2 \frac{x}{2} \\
 &= \frac{1}{2} (\tan^2 \frac{x}{2} + 1) = \frac{1}{2} (t^2 + 1) \\
 \therefore dx &= \frac{2dt}{t^2+1} \\
 x = \frac{\pi}{3}, t &= \frac{1}{\sqrt{3}}; x = \frac{\pi}{2}, t = 1
 \end{aligned}$$

## Question 2

$$\begin{aligned}
 (a) \quad 21 + 20i &= (x^2 - y^2) + 2xyi \\
 x^2 - y^2 &= 21 \quad \dots (1) \\
 2xy &= 20 \Rightarrow y = \frac{10}{x} \quad \dots (2) \\
 \text{sub (2) in (1)}: \quad x^2 - \frac{100}{x^2} &= 21
 \end{aligned}$$

$$\begin{aligned}
 x^4 - 21x^2 - 100 &= 0 \\
 (x^2 - 25)(x^2 + 4) &= 0 \\
 \therefore x &= \pm 5 \\
 x = 5 &\Rightarrow y = 2 \\
 x = -5 &\Rightarrow y = -2
 \end{aligned}$$

$$\therefore \sqrt{21+20i} = \underline{\underline{\pm(5+2i)}}$$



$$\begin{aligned}
 (c) \quad p &= (9+8i) - (3+5i) \\
 &= 6 + 3i
 \end{aligned}$$

$$\begin{aligned}
 (ii) \quad q &= i(6+3i) \\
 &= \underline{\underline{-3+6i}}
 \end{aligned}$$

$$\begin{aligned}
 (d) \quad z + \frac{1}{z} &= x+iy + \frac{1}{x+iy} \times \frac{x-iy}{x-iy} \\
 &= x+iy + \frac{x-iy}{x^2+y^2} \\
 &= \frac{x(x^2+y^2) + iy(x^2+y^2) + x - iy}{x^2+y^2} \\
 &= \underline{\underline{\frac{x(x^2+y^2+1) + iy(x^2+y^2-1)}{x^2+y^2}}}
 \end{aligned}$$

$$\operatorname{Im} \left( z + \frac{1}{z} \right) = \frac{y(x^2 + y^2 - 1)}{x^2 + y^2} = 0$$

$$\therefore y = 0 \quad \text{or} \quad x^2 + y^2 = 1, \quad x^2 + y^2 \neq 0$$

The locus of  $P$  is the unit circle and the  $x$ -axis except for the origin.

$$(2) \quad \sqrt{3} + i = 2 \operatorname{cis} \frac{\pi}{6}, \quad \sqrt{3} - i = 2 \operatorname{cis} \left( -\frac{\pi}{6} \right)$$

$$\therefore (\sqrt{3} + i)^{10} + (\sqrt{3} - i)^{10}$$

$$= (2 \operatorname{cis} \frac{\pi}{6})^{10} + (2 \operatorname{cis} (-\frac{\pi}{6}))^{10}$$

$$= 2^{10} \operatorname{cis} \frac{10\pi}{6} + 2^{10} \operatorname{cis} \left( -\frac{10\pi}{6} \right)$$

$$= 1024 \left\{ \operatorname{cis} \left( -\frac{\pi}{3} \right) + \operatorname{cis} \left( \frac{\pi}{3} \right) \right\}$$

$$= 1024 \left\{ \cos \left( -\frac{\pi}{3} \right) + i \sin \left( -\frac{\pi}{3} \right) + \cos \frac{\pi}{3} + i \sin \frac{\pi}{3} \right\}$$

$$= 1024 \left\{ \cos \frac{\pi}{3} - i \sin \frac{\pi}{3} + \cos \frac{\pi}{3} + i \sin \frac{\pi}{3} \right\}$$

$$= 1024 \left\{ \frac{1}{2} + \frac{1}{2} \right\}$$

$$= 1024 \quad \text{which is a rational number.}$$

### Question 3

$$(a) \quad x^3 - 3x^2 + 6x - 4 = 0 \quad \text{has real coefficients}$$

so  $1 - i\sqrt{3}$  is also a root.

$$\text{Let } \alpha = 1 + i\sqrt{3} \quad \text{and } \beta = 1 - i\sqrt{3}$$

$$\alpha + \beta + \gamma = 3$$

$$\therefore (1 + i\sqrt{3}) + (1 - i\sqrt{3}) + \gamma = 3$$

$$\Rightarrow \gamma = \underline{\underline{1}}$$

$$(b) \quad (i) \quad \text{Let } P(x) = (x-c)^m Q(x)$$

$$\therefore P'(x) = m(x-c)^{m-1} Q(x) + (x-c)^m Q'(x)$$

$$= (x-c)^{m-1} \{ m Q(x) + (x-c) Q'(x) \}$$

$\Rightarrow P'(x)$  has a root of multiplicity  $(m-1)$  at  $x=c$ .

$$(ii) \quad \text{Let } P(x) = 8x^3 + 4x^2 - 2x - 1$$

$$P'(x) = 24x^2 + 8x - 2$$

$$P'(x) = 0 \Rightarrow 12x^2 + 4x - 1 = 0$$

$$(6x-1)(2x+1) = 0$$

$$x = \frac{1}{6}, -\frac{1}{2}$$

$$P\left(\frac{1}{6}\right) = \frac{8}{216} + \frac{4}{36} - \frac{2}{6} - 1 \neq 0$$

$$P\left(-\frac{1}{2}\right) = -1 + 1 + 1 - 1 = 0$$

$\therefore x = -\frac{1}{2}$  is the double root.

$$(c) \quad \text{Put } y = \frac{1}{1-x}$$

$$1-x = \frac{1}{y}$$

$$x = 1 - \frac{1}{y}$$

$$x = \frac{y-1}{y}$$

$$\text{Req'd eq'n is } \left( \frac{y-1}{y} \right)^3 - \left( \frac{y-1}{y} \right) - 1 = 0$$

$$\therefore (y-1)^3 - y^2(y-1) - y^3 = 0$$

$$\therefore y^3 - 3y^2 + 3y - 1 - y^3 + y^2 - y^3 = 0$$

$$\therefore -y^3 - 2y^2 + 3y - 1 = 0$$

$$\text{i.e. } y^3 + 2y^2 - 3y + 1 = 0$$

$$\therefore \frac{1}{1-\alpha} = -2$$

(d) Let the roots be  $\alpha, \beta, \gamma$  where  $\alpha = \beta + \gamma$

$$\text{Sum of roots: } \alpha + \beta + \gamma = -a$$

$$\therefore 2\alpha = -a \Rightarrow \alpha = -\frac{a}{2}$$

$$\text{Sum of roots 2 at a time: } \alpha\beta + \alpha\gamma + \beta\gamma = b$$

$$\therefore \alpha(\beta + \gamma) + \beta\gamma = b$$

$$\therefore \left(-\frac{a}{2}\right)\left(-\frac{a}{2}\right) + \beta\gamma = b$$

$$\therefore \beta\gamma = -\frac{a^2}{4} + b \quad \dots (1)$$

$$\text{Product of roots: } \alpha\beta\gamma = -c$$

$$\therefore \beta\gamma = -\frac{c}{\alpha}$$

$$\therefore \beta\gamma = \frac{2c}{a} \quad \dots (2)$$

$$\text{from (1) \& (2): } -\frac{a^2}{4} + b = \frac{2c}{a}$$

$$\therefore -a^3 + 4ab = 8c$$

$$\therefore 0 = \underline{a^3 - 4ab + 8c}$$

### Question 4.

$$(a) (i) \phi^3 = 1, \phi \neq 1$$

$$\phi^3 - 1 = 0$$

$$(\phi - 1)(\phi^2 + \phi + 1) = 0$$

$$\therefore \phi^2 + \phi + 1 = 0 \quad (\phi \neq 1)$$

$$\therefore \underline{\phi^2 + \phi = -1}$$

$$(ii) (a-b)(a-\phi b)(a-\phi^2 b)$$

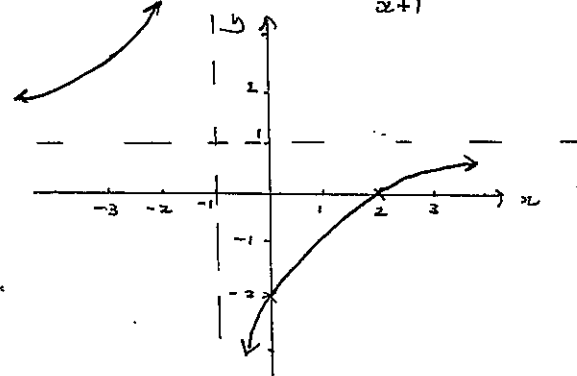
$$= (a-b)(a^2 - \phi^2 ab - \phi ab + \phi^3 b^2)$$

$$= (a-b)(a^2 + b^2 - ab(\phi^2 + \phi))$$

$$= (a-b)(a^2 + b^2 + ab)$$

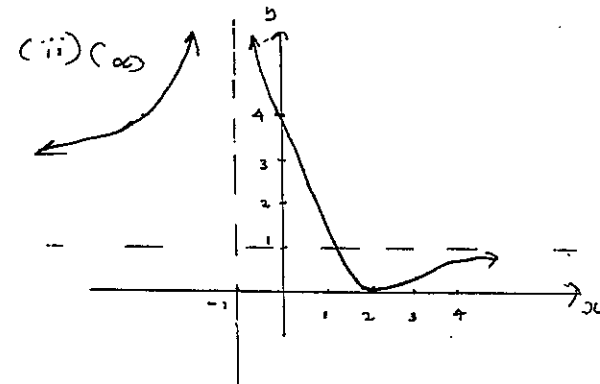
$$= \underline{a^3 - b^3}$$

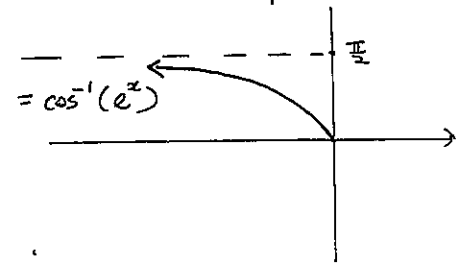
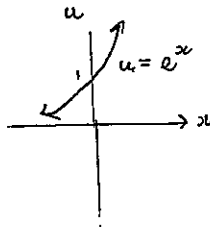
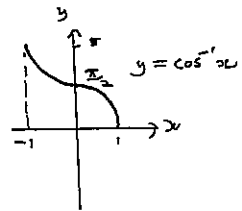
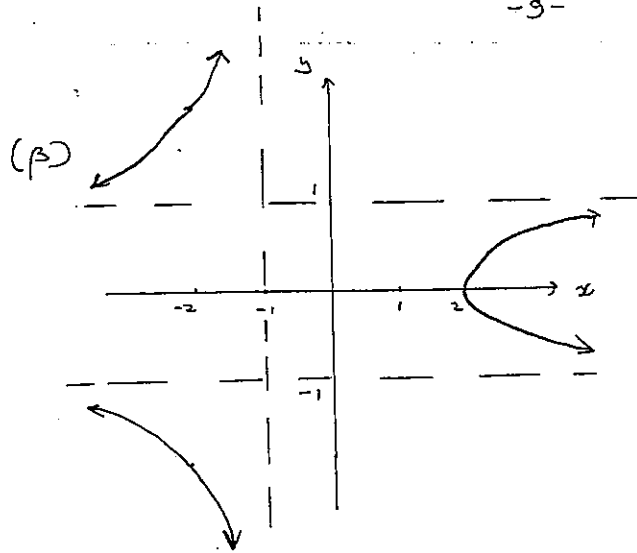
$$(b) (i) y = \frac{(x+1)-3}{x+1} = 1 - \frac{3}{x+1}$$



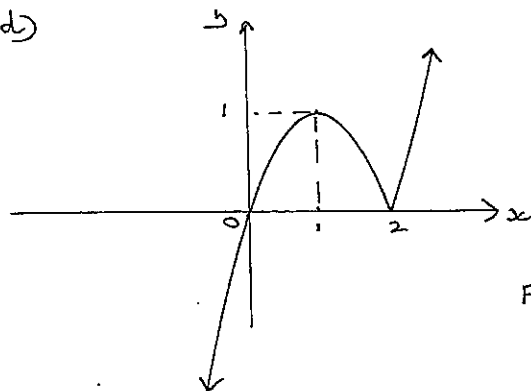
$$x=0, y=-2$$

$$y=0, x=2$$





(d)



when  $x=1$ ,  $y = 1 \times |1-2| = 1$

For what values of  $k$   
will  $y=k$  intersect  
 $y = x|x-2|$  in 3 places?

From the graph,  $0 < k < 1$ .

### Question 5

(a) (i)  $y = \frac{c^2}{x}$

$$\therefore \frac{dy}{dx} = -\frac{c^2}{x^2}$$

when  $x=ct$ ,  $\frac{dy}{dx} = -\frac{c^2}{c^2 t^2} = -\frac{1}{t^2} \Rightarrow m_N = t^2$

$\therefore$  Eq'n of normal:  $y - \frac{c}{t} = t^2(x - ct)$

$$ty - c = t^3 x - ct^4$$

$$ty + ct^4 = t^3 x + c$$

(ii) Tangent is  $x + t^2 y = 2ct$ .

when  $x=0$ ,  $t^2 y = 2ct$

$$y = \frac{2c}{t} \Rightarrow y \text{ is } (0, \frac{2c}{t})$$

when  $y=0$ ,  $x = 2ct \Rightarrow x \text{ is } (2ct, 0)$

for Q, sub  $y = -x$  in  $ty + ct^4 = t^3 x + c$ :

$$-xt + ct^4 = t^3 x + c$$

$$x(t^3 + t) = c(t^4 - 1)$$

$$x = \frac{c(t^4 - 1)(t^2 + 1)}{t(t^2 + 1)} = \frac{c(t^2 - 1)}{t}$$

sub  $x = \frac{c(t^2 - 1)}{t}$  in  $y = -x$

$$\therefore y = \frac{c(1 - t^2)}{t}$$

$$\Rightarrow Q \text{ is } \left( \frac{c(t^2 - 1)}{t}, \frac{c(1 - t^2)}{t} \right)$$

$$(ii) \text{ Midpoint of } PQ = \left( \frac{\frac{c}{t}(t^2+1+t^2-1)}{2}, \frac{\frac{c}{t}(t^2+1+1-t^2)}{2} \right)$$

$$= \left( \frac{c}{2t}(2t^2), \frac{c}{2t} \times 2 \right)$$

$$= \left( ct, \frac{c}{t} \right)$$

$$= R$$

$$\text{midpoint of } XY = \left( \frac{2ct+0}{2}, \frac{0+\frac{2c}{t}}{2} \right)$$

$$= \left( ct, \frac{c}{t} \right)$$

$$= R$$

$XY$  &  $PQ$  share the same midpoint, hence they must bisect each other.

$$(iv) \quad XY = \sqrt{(2ct)^2 + \left(\frac{2c}{t}\right)^2}$$

$$= \sqrt{4c^2t^2 + \frac{4c^2}{t^2}}$$

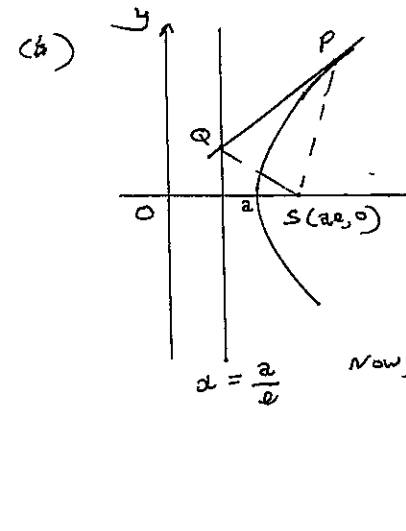
$$PQ = \sqrt{\left( \frac{c}{t}(t^2+1) - \frac{c}{t}(t^2-1) \right)^2 + \left( \frac{c}{t}(t^2+1) - \frac{c}{t}(1-t^2) \right)^2}$$

$$= \sqrt{\frac{4c^2}{t^2} + \frac{c^2}{t^2} \times 4t^4}$$

$$\therefore PQ = \sqrt{\frac{4c^2}{t^2} + 4c^2t^2} = XY$$

(v)  $\therefore$  The diagonals bisect each other at right angles and are equal in length,  $XQYP$  is a square.

-12-

(b) 

Coordinates of Q: when  $x = \frac{a}{e}$ ,  $\frac{\sec \theta}{e} - \frac{y \tan \theta}{b} = 1$

$\therefore b \sec \theta - ay \tan \theta = be$

$\Rightarrow y = \frac{b(\sec \theta - e)}{a \tan \theta}$

Now,  $m_{QS} = \frac{\frac{b(\sec \theta - e)}{a \tan \theta}}{\frac{a}{e} - ae}$

$= \frac{b(\sec \theta - e)}{a \tan \theta (1 - e^2)}$

and  $m_{PS} = \frac{b \tan \theta}{a \sec \theta - ae} = \frac{b \tan \theta}{a(\sec \theta - e)}$

$$m_{PS} \times m_{QS} = \frac{b(\sec \theta - e)}{a \tan \theta (1 - e^2)} \times \frac{b \tan \theta}{a(\sec \theta - e)}$$

$$= \frac{b^2}{a^2(1 - e^2)}$$

i.e.  $m_{PS} \times m_{QS} = -1$  as  $b^2 = a^2(e^2 - 1)$

$\Rightarrow PS \perp QS$

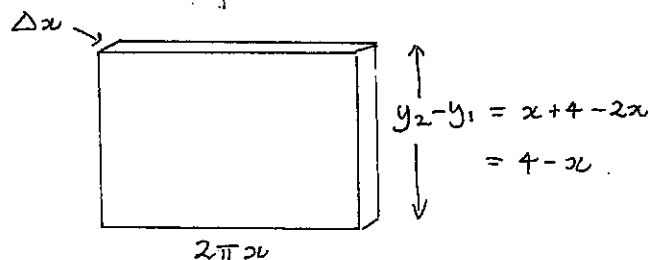
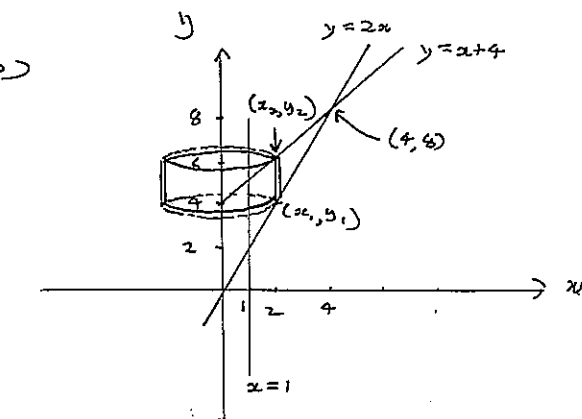
Question 6

(2)  $\cos \theta = \frac{x-1}{3}$ ,  $\sin \theta = \frac{y+2}{2}$

$\therefore \frac{(x-1)^2}{9} + \frac{(y+2)^2}{4} = \cos^2 \theta + \sin^2 \theta$

$\therefore \frac{(x-1)^2}{9} + \frac{(y+2)^2}{4} = 1$

(b)  $y=2x$   $y=x+4$   $\Rightarrow x=4, y=8$



$\Delta V \doteq 2\pi x (4-x) \Delta x$

OR  
 $\Delta V = \pi [(x+\Delta x)^2 - x^2] [4-x]$   
 $= \pi [2x\Delta x + (\Delta x)^2] [4-x]$

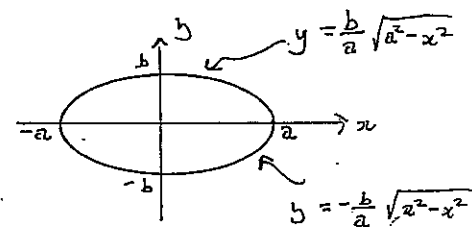
$V = \lim_{\Delta x \rightarrow 0} \sum_{x=1}^4 \pi [2x\Delta x + (\Delta x)^2] [4-x]$   
 $= \lim_{\Delta x \rightarrow 0} \sum_{x=1}^4 2\pi (x + \frac{\Delta x}{2}) (4-x) \Delta x$

$V = 2\pi \int_{x=1}^{x=4} x(4-x) dx$   
 $= 2\pi \int_1^4 (4x - x^2) dx$   
 $= 2\pi \left[ 2x^2 - \frac{x^3}{3} \right]_1^4$   
 $= 2\pi \left[ (32 - \frac{64}{3}) - (2 - \frac{1}{3}) \right]$   
 $= 18\pi u^3$

(c)

(i)  $\int_0^a \sqrt{a^2 - x^2} dx$  calculates the area of the 1st quadrant of a circle centred at the origin with radius  $a$

$\therefore \int_0^a \sqrt{a^2 - x^2} dx = \frac{1}{4} \times \pi \times a^2$   
 $= \frac{\pi a^2}{4}$

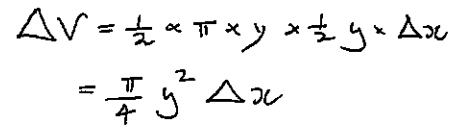


(ii)  $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

$y^2 = b^2 (1 - \frac{x^2}{a^2})$   
 $= \frac{b^2}{a^2} (a^2 - x^2)$

$y = \pm \frac{b}{a} \sqrt{a^2 - x^2}$

$A = \frac{2b}{a} \int_{-a}^a \sqrt{a^2 - x^2} dx$   
 $= \frac{4b}{a} \int_0^a \sqrt{a^2 - x^2} dx$   
 $= \frac{4b}{a} \times \frac{\pi a^2}{4}$   
 $= \pi ab u^2$



$$\therefore V = \frac{\pi}{4} \int_{-r}^r (r^2 - x^2) dx$$

$$= \frac{\pi}{2} \int_0^r (r^2 - u^2) du$$

$$= \frac{\pi}{2} \left[ x^2 - \frac{x^3}{3} \right]_0^1$$

$$= \frac{\pi}{2} \left\{ r^3 - \frac{r^3}{3} \right\}$$

$$= \frac{\pi}{2} \times \frac{2r^3}{3}$$

$$= \frac{\pi r^3}{3} \text{ units}^3$$

(b) Prob. of 2 girls  
& 2 boys in  
1 family  $\left. \vphantom{\begin{matrix} \text{Prob. of 2 girls} \\ \text{\& 2 boys in} \\ \text{1 family} \end{matrix}} \right\} = {}^4C_2 \left(\frac{1}{2}\right)^2 \left(\frac{1}{2}\right)^2$   
 $= \frac{3}{8}$

$$\begin{aligned} \text{Reqd prob.} &= {}^4C_2 \left(\frac{3}{8}\right)^2 \left(\frac{5}{8}\right)^2 \\ &= \frac{6 \times 9 \times 25}{8^4} \\ &= \frac{675}{2048} \end{aligned}$$

(c) (i) Req<sup>d</sup> no. =  ${}^5C_2 \times 3!$   
 $= 10 \times 6$   
 $= \underline{\underline{60}}$

(ii) No. with no E's =  ${}^5C_3 \times 3!$  or  $5 \times 4 \times 3 = 60$   
 $= 10 \times 6$   
 $= 60$

No. with 1 E = 60 from (i)  
No. with 2 E's =  ${}^5C_1 \times 3$   
 $= 5 \times 3$   
 $= 15$

No. with 3 E's = 1

Total no. =  $60 + 60 + 15 + 1$   
= 136

(d)

|        |       |            |
|--------|-------|------------|
| 8:20pm | 13.6m | $x = 3.2$  |
|        | 12.3m | $x = 1.9$  |
|        | 10.4m | $x = 0$    |
| 2:05pm | 7.2m  | $x = -3.2$ |

Let  $x = A \cos(\omega t + \alpha)$   
where  $t$  is time in minutes.

$$\therefore x = 3.2 \cos\left(\frac{\pi t}{375} + \alpha\right)$$

When  $t=0$ ,  $x = -3.2$

$$-3.2 = 3.2 \cos \alpha$$

$$\cos \alpha = -1$$

$$\alpha = \pi$$

$$\therefore x = 3.2 \cos\left(\frac{\pi t}{375} + \pi\right)$$

$$= -3.2 \cos\left(\frac{\pi t}{375}\right)$$

When  $x = 1.9$ ,  $1.9 = -3.2 \cos\left(\frac{\pi t}{375}\right)$

$$\cos\left(\frac{\pi t}{375}\right) = -\frac{1.9}{3.2}$$

$$\frac{\pi t}{375} = \dots, \underbrace{\cos^{-1}\left(\frac{1.9}{3.2}\right)}_X, \underbrace{\pi - \cos^{-1}\left(\frac{1.9}{3.2}\right)}_{\uparrow}, \underbrace{\pi + \cos^{-1}\left(\frac{1.9}{3.2}\right)}_{\uparrow}, \dots$$

$$t = \dots, 263.38\dots, 486.61\dots$$

$$t = 264 \text{ (rounding up)}, 486 \text{ (rounding down)}$$

$\therefore$  He should leave between 6:29 pm and 10:11 pm.

OR

|           |       |            |
|-----------|-------|------------|
| 8:20pm    | 13.6  | $x = 3.2$  |
|           | 12.3m | $x = 1.9$  |
| 5:12:30pm | 10.4  | $x = 0$    |
| 2:05pm    | 7.2m  | $x = -3.2$ |

Let  $x = A \sin \omega t$ ,  
taking  $t=0$  when  $x=0$ .

$$A = 3.2, \quad \omega = \frac{2\pi}{T}, \quad T = 375 \times 2 = 750$$

$$= \frac{\pi}{375}$$

$$\therefore x = 3.2 \sin\left(\frac{\pi t}{375}\right)$$

When  $x = 1.9$ ,

$$1.9 = 3.2 \sin\left(\frac{\pi t}{375}\right)$$

$$\frac{\pi t}{375} = \sin^{-1}\left(\frac{1.9}{3.2}\right), \pi - \sin^{-1}\left(\frac{1.9}{3.2}\right), \dots$$

$$t = 75.88\dots, 299.11\dots$$

$\therefore$  He should leave between 6:29 pm and 10:11 pm.

$\uparrow$   
rounded up

$\uparrow$   
rounded down

(a) (i)  $(\cos \theta + i \sin \theta)^3$   
 $= \cos^3 \theta + 3 \cos^2 \theta i \sin \theta + 3 \cos \theta (i \sin \theta)^2 + (i \sin \theta)^3$   
 By De Moivre's theorem  $(\cos \theta + i \sin \theta)^3 = \cos 3\theta + i \sin 3\theta$   
 $\therefore \cos 3\theta + i \sin 3\theta = \cos^3 \theta - 3 \cos \theta \sin^2 \theta + i(3 \cos^2 \theta \sin \theta - \sin^3 \theta)$   
 Equating real parts:  
 $\cos 3\theta = \cos^3 \theta - 3 \cos \theta (1 - \cos^2 \theta)$   
 $= 4 \cos^3 \theta - 3 \cos \theta$

(ii)  $8x^3 - 6x + 1 = 0$   
 $8x^3 - 6x = -1$   
 $4x^3 - 3x = -\frac{1}{2}$   
 $4 \cos^3 \theta - 3 \cos \theta = -\frac{1}{2}$ , where  $x = \cos \theta$   
 $\therefore \cos 3\theta = -\frac{1}{2}$  { from (i) }

$$3\theta = \frac{2\pi}{3}, \frac{4\pi}{3}, \frac{8\pi}{3}$$

$$\theta = \frac{2\pi}{9}, \frac{4\pi}{9}, \frac{8\pi}{9}$$

$$\therefore x = \cos \frac{2\pi}{9}, \cos \frac{4\pi}{9}, \cos \frac{8\pi}{9}$$

(iii) Product of roots  $= -\frac{1}{8}$   
 $\therefore \cos \frac{2\pi}{9} \cos \frac{4\pi}{9} \cos \frac{8\pi}{9} = -\frac{1}{8}$   
 $\cos \frac{2\pi}{9} \cos \frac{4\pi}{9} (-\cos \frac{\pi}{9}) = -\frac{1}{8}$   
 $\cos \frac{\pi}{9} \cos \frac{2\pi}{9} \cos \frac{4\pi}{9} = \frac{1}{8}$   
 $\frac{1}{\cos \frac{\pi}{9} \cos \frac{2\pi}{9} \cos \frac{4\pi}{9}} = 8$   
 $\therefore \sec \frac{\pi}{9} \sec \frac{2\pi}{9} \sec \frac{4\pi}{9} = 8$

(b) (i)  $I_n = \int_0^{\frac{\pi}{2}} \sin^n x \, dx$   
 $= \int_0^{\frac{\pi}{2}} \sin^{n-1} x \, d(-\cos x)$   
 $= \left[ -\cos x \sin^{n-1} x \right]_0^{\frac{\pi}{2}} + \int_0^{\frac{\pi}{2}} \cos x \, d(\sin^{n-1} x)$   
 $= 0 + \int_0^{\frac{\pi}{2}} \cos x (n-1) \sin^{n-2} x \cos x \, dx$   
 $= (n-1) \int_0^{\frac{\pi}{2}} \sin^{n-2} x (1 - \sin^2 x) \, dx$   
 $= (n-1) \left\{ \int_0^{\frac{\pi}{2}} \sin^{n-2} x \, dx - \int_0^{\frac{\pi}{2}} \sin^n x \, dx \right\}$   
 $= (n-1) \{ I_{n-2} - I_n \}$

$$I_n + (n-1) I_n = (n-1) I_{n-2}$$

$$n I_n = (n-1) I_{n-2}$$

$$\therefore I_n = \frac{n-1}{n} I_{n-2}$$

$$\begin{aligned}
 \text{(ii)} \quad I_1 &= \int_0^{\frac{\pi}{2}} \sin x \, dx \\
 &= [-\cos x]_0^{\frac{\pi}{2}} \\
 &= (-\cos \frac{\pi}{2}) - (-\cos 0) \\
 &= 1
 \end{aligned}$$

$$I_3 = \frac{2}{3} \times 1 = \frac{2}{3}$$

$$I_5 = \frac{4}{5} \times I_3 = \frac{2}{3} \times \frac{4}{5}$$

$$I_7 = \frac{2 \times 4 \times 6}{3 \times 5 \times 7}$$

$$I_9 = \frac{2 \times 4 \times 6 \times 8}{3 \times 5 \times 7 \times 9}$$

$$\therefore I_n = \frac{2 \times 4 \times 6 \times 8 \times \dots \times (n-1)}{1 \times 3 \times 5 \times 7 \times 9 \times \dots \times n} \quad \text{where } n \text{ is an odd positive integer}$$

$$= \frac{2 \times 4 \times 6 \times 8 \times \dots \times (n-1)}{1 \times 3 \times 5 \times 7 \times 9 \times \dots \times n} \times \frac{2 \times 4 \times 6 \times 8 \times \dots \times (n-1)}{2 \times 4 \times 6 \times 8 \times \dots \times (n-1)}$$

$$= \frac{\{ 2^{(n-1)/2} (1 \times 2 \times 3 \times \dots \times \frac{n-1}{2}) \}^2}{n!}$$

$$= \frac{2^{n-1} \{ (\frac{n-1}{2})! \}^2}{n!}$$